

# Light Field Transmission Fusing Image Super-Resolution and Selective Quality Patterns

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## ABSTRACT

Light field imaging has notably advanced immersive experiences and virtual reality applications. However, this progress is accompanied by a considerable increase in data volume, posing challenges in storage and transmission efficiency. In this study, we propose a novel framework for light field image storage and transmission. The approach leverages the image super-resolution and pseudo video coding techniques to process LF images as video sequences. Meanwhile, a series of selective transmission patterns tailored to human visual perception are developed. The experimental results show that the approach optimizes the transmission efficiency while preserving similar LF image quality.

**Index Terms:** Light field image—Image super-resolution—Intelligent transmission—HEVC

## 1 INTRODUCTION

Light Field (LF) technology [7] captures both light intensity and direction, enabling post-capture adjustments like refocusing and perspective changes. The synergy between LF and virtual reality (VR) is pivotal, wherein the simulation and reproduction of the propagation of light play a crucial role in crafting more realistic and immersive virtual experiences for users [7].

However, the complex data structure of LF inherently results in considerably larger data volumes compared to 2D images, prompting the need for intelligent LF data transmission. A significant challenge in this domain is the transmission efficiency given its voluminous nature. The conventional transmission methodology delivers the entire LF image at a uniformly high resolution [8], inevitably imposing a strict bandwidth requirement.

In light of the aforementioned challenge, we present Super-Resolution-based Selective Transmission Network (SRST-Net), a framework seamlessly integrating image super-resolution, a technique in image processing that aims to enhance the resolution and detail of an image beyond its original size or quality [1], into the intelligent transmission of light field data. The framework transmits only a low-resolution LF images and the residues against the super-resolved images. The LF sub-views are arranged into a pseudo video sequence that can be encoded with HEVC [5], as depicted in Figure 1. Moreover, to better cater to the perceptual differences in various regions of LF scenes as perceived by human eyes, a series of transmission patterns are proposed, including zoom-center, cross-in, and cube-center (detailed in Section 2.2). The experimental results demonstrate that the proposed approach not only reduces the amount of data transmitted but also maintains a high-quality visual experience aligning with human visual perception.

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This work was supported in part by the Beijing Natural Science Foundation under Grant 4222003 and the National Natural Science Foundation of China under Grant 62177001.

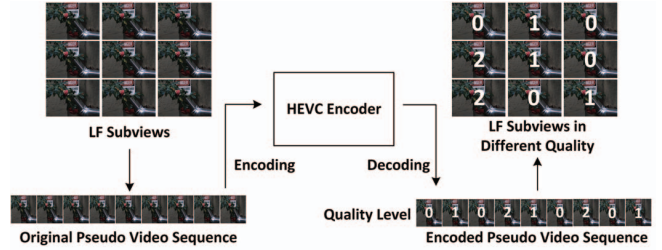


Figure 1: Block diagram showing the LF data transmission process.

## 2 METHODOLOGY

### 2.1 Image SR-based Intelligent Transmission

Due to the characteristics of LF images, transmitting and storing LF data demands intensive resources. To address this issue, we propose an intelligent LF image SR-based transmission approach, SRST-Net, as shown in Figure 2.

In the LF image encoder, downsampling ( $D$ ) is conducted to reduce the image resolution by a factor of  $\alpha$ , followed by an image super-resolution operation ( $F$ ). The super-resolution is applied to each individual subview of the LF by using the ESRGAN method [6], yielding a high-resolution LF image. Then the high-resolution LF image is subtracted from the original LF image, producing the residual information  $R_i$ . Finally, the low-resolution LF data  $I_{LR}$  and residual information  $R_i$  are encoded, described by the following equations:

$$I_{LR} = D(I_{HR}), \quad I_{HR} \in \mathbb{R}^{U \times V \times W \times H},$$

$$R_i = I_{HR} - F(I_{LR}), \quad I_{LR} \in \mathbb{R}^{U \times V \times W / \alpha \times H / \alpha},$$

where  $(U, V)$  and  $(W, H)$  denote the angular and spatial subspaces of the LR image, with  $\alpha$  as the downsampling factor.

In the decoder, the transmitted information is decoded to retrieve low-resolution LF data  $I_{LR}$  and residual information  $R_i$ . With  $F$ , the low-resolution LF image is super-resolved to the high-resolution LF image, combined with the decoded residual information  $R_i$ , yielding high-quality LF data approximating the original LF data. This process can be described by the following equation:

$$I_{HR} = F(I_{LR}) + R_i, \quad I_{LR} \in \mathbb{R}^{U \times V \times W / \alpha \times H / \alpha},$$

$$I_{HR} \in \mathbb{R}^{U \times V \times W \times H},$$

### 2.2 Selective Quality-level Transmission Patterns

Traditional methods directly transmit the entire light field with equal quality, undoubtedly leading to bandwidth waste. In contrast, we employ three selective transmission patterns, including zoom-center, cross-in, and cube-center, as illustrated in Figure 3.

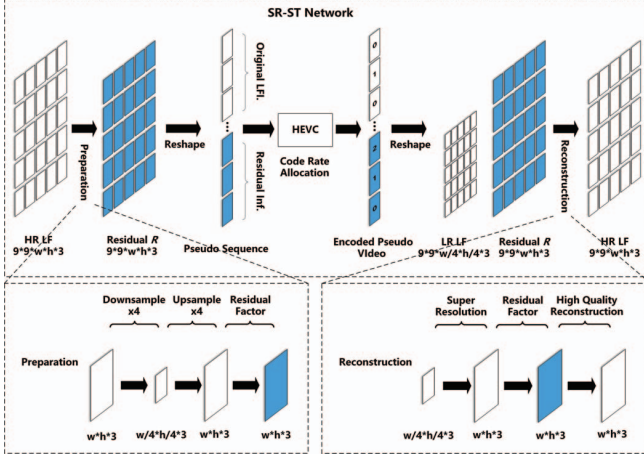


Figure 2: Architecture of *SRST-Net*, the input is a 9×9 LF image, and the output is the coded pseudo video consisting of LF subviews in different quality levels.

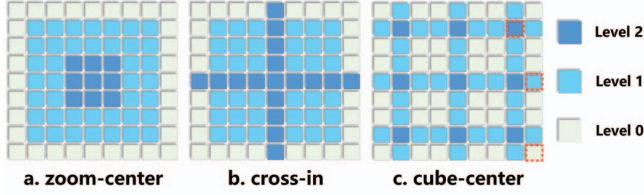


Figure 3: The proposed three selective transmission patterns, among which quality levels are represented by different colors.

First, the zoom-center pattern prioritizes high-quality transmission for the central light field subviews and using lower resolutions for those farther from the center. Second, the cross-in pattern selectively transmits high-quality subviews along the horizontal and vertical axes, aligning with the human eye’s visual characteristics. Finally, the cube-center pattern employs a grid-based selective transmission that optimizes bandwidth by dividing the light field into local grid regions. For each region, the subview at the grid center undergoes high-quality transmission, while quality is reduced for subviews at the grid corners. This assumes that central portions contain crucial details for viewer perception.

During the encoding process, we obtain the specific QP value for HEVC encoding through the following formula:

$$QP = a \cdot e^{-bX},$$

where  $a$  and  $b$  are experimentally set to 24 to 0.3, respectively. We employ three quality levels (0, 1, and 2), corresponding to QP values of 24 (level 0), 18 (level 1), and 13 (level 2) of the HEVC encoding. The strategy recognizes user focus on the central image, allowing lower resolution for the side views of the LF to save bandwidth during transmission.

### 3 EXPERIMENTAL RESULTS

In our experiments, 12 groups of 12 light field images each were sourced from DUTLF-V2 [2]. We randomly selected four light field datasets (SingOJ, SimBG, SimEG, TransOJ) from four scene categories, conducting the entire experimental process with a image super-resolution method [6]. For HEVC encoding, ffmpeg version 6.1 was used to encode the LF subviews into pseudo videos. In the literature, the quality of a LF image can be objectively evaluated

Table 1: The performance evaluation in data size (MB) / quality score (by the LFACon metric [4])

| Scenes            | SingOJ      | SimBG       | SimEG       | TransOJ     |
|-------------------|-------------|-------------|-------------|-------------|
| Original encoding | 0.58 / 0.69 | 0.75 / 0.71 | 0.97 / 0.67 | 1.38 / 0.92 |
| SR only           | 0.19 / 0.53 | 0.26 / 0.78 | 0.29 / 0.66 | 0.46 / 0.78 |
| SR+zoom-center    | 0.12 / 0.50 | 0.17 / 0.37 | 0.19 / 0.74 | 0.27 / 0.93 |
| SR+cross-in       | 0.17 / 0.47 | 0.23 / 0.39 | 0.26 / 0.63 | 0.47 / 0.87 |
| SR+cube-center    | 0.16 / 0.50 | 0.19 / 0.37 | 0.22 / 0.70 | 0.39 / 0.77 |

with LF image quality assessment metrics [3] [4]. In this study, the state-of-the-art light field quality assessment metric, LFACon [4], is employed to evaluate the quality of the transmitted LF images. As observed from the presented results in Table 1, our SRST-Net (SR-based transmission combined with our selective transmission patterns denoted by “SR only”, “SR+zoom-center”, “SR+cross-in”, and “SR+cube-center”) demonstrates more efficient transmission of LF data. The size of LF data is substantially reduced while maintaining reasonable quality scores. In certain instances, the quality scores achieved by SRST-Net even surpass the original encoding baseline (LF data encoded without super-resolution and selective patterns, QP=13), which typically necessitates higher storage and bandwidth. However, it is important to note that this phenomenon might be attributed to potential biases in the LFACon metric. We will further investigate this problem and employ more LF quality assessment metrics in our future work.

### 4 CONCLUSION

In this poster, we introduce our SRST-Net that seamlessly integrates image super-resolution into the intelligent transmission process of light field data. Our approach incorporates different transmission patterns for LF images, allowing for selective transmission by prioritizing regions of interest to users. This results in improved high-quality restoration in crucial areas, as evidenced by the experimental results. The intelligent downsampling of the less significant regions from transmission through the selective transmission mechanism effectively reduces the bandwidth requirement, further amplifying the overall efficiency of intelligent transmission for LF data.

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